BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

# Endocrine anatomy

The written notes for the endocrine lecture: the major glands and where each sits, the pituitary and its two lobes, the thyroid and parathyroid glands, the adrenal zones, and the pancreas, gonads, and other organs with endocrine tissue. Module 4 competencies 128 to 132, with nine worked questions.

**To print:** these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

**How to use these notes.** Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

## Before the competencies: what an endocrine gland is

**In your course notes:** Module 4 packet, Chapter 6 introduction (page 52).

**Dr. Rennie's video for this part:** Endocrine system concept video, chapter Endocrine system overview and purpose.

The endocrine system is a scattered set of glands and tissues that release hormones, chemical messengers that travel in the blood. Unlike most organ systems, most of its glands are not physically connected to each other. They sit in the head, neck, thorax, abdomen, pelvis, and scrotum, and the blood is what links them.

An endocrine gland is a ductless gland. Its cells release their product into the surrounding interstitial fluid, and from there it enters the blood. That is why endocrine glands have such a rich supply of capillaries. An exocrine gland is different: it has a duct that carries its product onto a surface or into a cavity, the way the salivary glands empty into the mouth. Some organs, such as the pancreas, have both kinds of tissue.

**Endocrine system**
: The glands and tissues that release hormones into the blood to signal distant cells.

**Hormone**
: A chemical messenger released by an endocrine cell that travels in the blood to target cells.

**Endocrine gland**
: A ductless gland that releases its product into the surrounding fluid and the blood.

**Exocrine gland**
: A gland that releases its product through a duct onto a surface or into a cavity. Not part of the endocrine system.

**Target cell**
: A cell that carries the receptor for a given hormone and so responds to it.

- Ductless and into the blood means endocrine. Through a duct and onto a surface means exocrine.
- Because hormones travel in the blood, an endocrine gland can be far from the cells it controls.

## Competency 128. Major endocrine glands and locations

**In your course notes:** Module 4 packet, Chapter 6, competency 128 (page 52).

**Dr. Rennie's video for this part:** Endocrine system concept video, chapter Major glands and their roles.

**You should be able to:** Name the major endocrine glands and locate each one in the body from the hypothalamus to the gonads.

Learn the major glands from the top of the body down. Three are in the brain or hang from it: the hypothalamus, the pituitary gland, and the pineal gland. Two are in the neck: the thyroid gland and the parathyroid glands behind it. The thymus is in the thorax. The adrenal glands and the pancreas are in the abdomen, and the endocrine tissue of the pancreas is the pancreatic islets. The gonads are the ovaries in the female pelvis and the testes in the male scrotum.

### Where each gland sits

| Gland | Body region | Location |
|---|---|---|
| **Hypothalamus** | Brain | The lowest part of the diencephalon, below the thalamus. It forms the floor and lower walls of the third ventricle. It links the nervous and endocrine systems and controls the pituitary |
| **Pituitary gland** | Brain | A pea-sized gland in the sella turcica of the sphenoid bone, hanging below the hypothalamus by the infundibulum |
| **Pineal gland** | Brain | A small, cone-shaped gland in the epithalamus, projecting from the posterior roof of the third ventricle |
| **Thyroid gland** | Neck | A butterfly-shaped gland in the anterior neck just below the larynx. Its two lobes lie beside the trachea and its isthmus crosses in front of it. The largest purely endocrine gland |
| **Parathyroid glands** | Neck | Usually four tiny glands on the posterior surface of the thyroid lobes, two on each side |
| **Thymus** | Thorax | In the mediastinum, behind the sternum and above the heart. Large in childhood, shrinks after puberty |
| **Adrenal glands** | Abdomen | Also called the suprarenal glands. One on the superior pole of each kidney, on the posterior abdominal wall behind the peritoneum |
| **Pancreatic islets** | Abdomen | Small clusters of endocrine cells scattered through the pancreas, which lies behind the stomach |
| **Ovaries** | Pelvis | In the pelvic cavity, one on each side of the uterus |
| **Testes** | Scrotum | In the scrotum, outside the abdominopelvic cavity |

- Memory tip: learn the glands from top to bottom by body region. Brain: hypothalamus, pituitary, pineal. Neck: thyroid and parathyroids. Chest: thymus. Abdomen: adrenals and pancreas. Pelvis: ovaries, or testes in the scrotum.
- The pineal gland sits at the back of the third ventricle and the pituitary hangs below the hypothalamus, in front and below. They are not in the same spot.
- Four of the glands come in pairs or more: the parathyroids (usually four, not two), the adrenals, the ovaries, and the testes. The pancreatic islets are many small clusters inside one pancreas.
- The thyroid is the largest gland in the body that is purely endocrine. The thymus is large in childhood and shrinks after puberty.

### What this means for patients

- The pituitary gland sits in the bony sella turcica, so a growing pituitary tumor usually expands upward out of the sella. Directly above it is the optic chiasm, and pressure there can cause a loss of peripheral vision. That vision change can be the first sign that a gland the size of a pea has enlarged.
- The thyroid gland lies just under the skin of the anterior neck, so an enlarged thyroid, a goiter, is visible and can be felt. The same superficial position lets a clinician palpate the gland and makes thyroid surgery accessible, though the tiny parathyroid glands on its back surface must be carefully spared.
- Each adrenal gland rests on the superior pole of a kidney. Imaging ordered for the kidney also captures the adrenal, which is why an adrenal mass is sometimes found by accident.

## Competency 129. Pituitary gross anatomy and lobes

**In your course notes:** Module 4 packet, Chapter 6, competency 129 (page 54).

**Dr. Rennie's video for this part:** Endocrine system concept video, chapter Hypothalamus and pituitary control.

**You should be able to:** Describe the pituitary in the sella turcica, its infundibulum link to the hypothalamus, and compare the anterior and posterior lobes by tissue type.

The pituitary gland, also called the hypophysis, is about the size of a pea. It sits in the sella turcica, a saddle-shaped pocket in the body of the sphenoid bone in the middle of the base of the skull. The bone forms the floor and the front and back walls of this pocket and protects the gland. The optic chiasm lies just above and in front of it.

The pituitary hangs from the hypothalamus by a slender stalk called the infundibulum. The infundibulum is the physical link between the brain and the gland. Nerve fibers from the hypothalamus run down through it, and small blood vessels run down along it.

**Hypophysis**
: Another name for the pituitary gland.

**Sella turcica**
: The saddle-shaped pocket of the sphenoid bone that cradles and protects the pituitary.

**Infundibulum**
: The slender stalk that connects the pituitary to the hypothalamus above it.

**Hypothalamo-hypophyseal tract**
: The bundle of nerve fibers that runs from the hypothalamus down through the infundibulum to the posterior lobe.

**Hypophyseal portal system**
: The blood vessels that carry hypothalamic signals directly to the anterior lobe.

### The two lobes

The pituitary has two lobes that develop from two different tissues in the embryo, so it is really two glands joined together. The anterior lobe, or adenohypophysis, grows up from the roof of the embryonic mouth. It is made of glandular epithelial tissue. Its cells are arranged in cords and clusters with wide capillaries running between them. The anterior lobe is the larger of the two.

The posterior lobe, or neurohypophysis, grows down from the floor of the brain. It is made of nervous tissue and is an extension of the hypothalamus itself. It contains the axons of neurons whose cell bodies sit in the hypothalamus, along with supporting glial cells called pituicytes. The posterior lobe makes no hormones of its own. It stores and releases what the hypothalamic neurons make and send down their axons.

Because the tissues differ, each lobe connects to the hypothalamus in a different way. The posterior lobe connects by nerve fibers, the hypothalamo-hypophyseal tract, which runs down through the infundibulum. The anterior lobe has no nerve link to the hypothalamus. It connects through blood vessels, the hypophyseal portal system, which has three parts in order:

1. Primary capillary plexus: a capillary bed at the base of the hypothalamus, where the infundibulum begins, picks up hypothalamic signals.
2. Hypophyseal portal veins: small veins carry the blood down along the infundibulum.
3. Secondary capillary plexus: a second capillary bed in the anterior lobe delivers the signals to the glandular cells.

| Feature | Anterior lobe | Posterior lobe |
|---|---|---|
| **Other name** | Adenohypophysis | Neurohypophysis |
| **Tissue type** | Glandular epithelial tissue | Nervous tissue, an extension of the hypothalamus |
| **Develops from** | An upgrowth of the roof of the embryonic mouth | A downgrowth of the floor of the brain |
| **Link to the hypothalamus** | Blood vessels, the hypophyseal portal system | Nerve fibers, the hypothalamo-hypophyseal tract, through the infundibulum |
| **On a slide** | Densely packed gland cells, many stained pink or blue, in cords with capillaries between them | Pale, fibrous-looking tissue of axons and scattered pituicyte nuclei |

- Memory tip: "neuro" means nervous tissue and "adeno" means gland. The posterior lobe is nervous tissue, so it connects by nerve fibers. The anterior lobe is glandular epithelium, so it connects by blood vessels.
- The tissue type predicts the link. Nervous tissue links by nerve fibers; glandular tissue links by blood vessels. The nerve fibers go to the posterior lobe and the blood vessels go to the anterior lobe, never the other way around.
- A pituitary tumor grows up and out of the sella turcica, because bone closes the floor and walls, and the first thing it presses on is the optic chiasm.

## Competency 130. Thyroid and parathyroid anatomy

**In your course notes:** Module 4 packet, Chapter 6, competency 130 (page 55).

**Dr. Rennie's video for this part:** Endocrine system concept video, chapter Thyroid and calcium regulation.

**You should be able to:** Identify the thyroid lobes, isthmus, follicles, colloid, follicular and parafollicular cells, and the parathyroid chief and oxyphil cells.

### The thyroid gland, gross anatomy

The thyroid gland is a butterfly-shaped gland in the anterior neck, just below the thyroid cartilage of the larynx. It has a right lobe and a left lobe. Each lobe lies against the side of the larynx and trachea. The two lobes are joined across the midline by a narrow bridge of thyroid tissue called the isthmus, which crosses in front of the upper rings of the trachea, just below the cricoid cartilage. A thin connective tissue capsule covers the gland. The thyroid is the largest gland in the body that is purely endocrine, and it has a very rich blood supply.

### The thyroid gland, on the slide

Under the microscope, the thyroid is made of thousands of tiny hollow spheres called thyroid follicles. Each follicle is a ring of cells around a center filled with colloid. Colloid is a thick, protein-rich fluid. It stains pink and fills the center of each follicle.

The wall of each follicle is a single layer of follicular cells. They form a simple cuboidal epithelium that sits around the colloid. Follicular cells are the main hormone-making cells of the thyroid. Between the follicles, or tucked into the outer edge of a follicle wall, are the parafollicular cells, also called C cells. They are larger and paler than follicular cells, and they do not touch the colloid. They occur alone or in small groups.

| Structure | What it is | How to find it |
|---|---|---|
| **Right and left lobes** | The two lateral parts of the gland | On each side of the larynx and trachea, below the thyroid cartilage |
| **Isthmus** | The bridge of thyroid tissue that connects the two lobes | Crossing the front of the upper trachea in the midline |
| **Thyroid follicles** | Microscopic hollow spheres that make up the gland | Rings of cells packed together across the slide |
| **Colloid** | The protein-rich fluid stored in the center of each follicle | Pink material filling each ring |
| **Follicular cells** | The simple cuboidal cells forming the wall of a follicle | The single layer of cells lining each ring of colloid |
| **Parafollicular (C) cells** | Endocrine cells between the follicles | Larger, pale cells between follicles or at the outer edge of a follicle wall, not touching the colloid |

- The lobes and isthmus are things you identify on the model or the cadaver. Follicles, colloid, and C cells are too small to see by eye, so those are slide structures only.
- Colloid fills the inside of the follicle. The C cells sit between follicles or at the outer edge of the wall, and they never touch the colloid.

### The parathyroid glands

The parathyroid glands are small, round glands, each about the size of a grain of rice. Most people have four, a superior and an inferior gland on the posterior surface of each thyroid lobe. Each one has its own thin capsule that separates it from the thyroid tissue. The number and exact position vary from person to person.

Parathyroid tissue has no follicles and no colloid. Its cells are packed in cords and clusters with capillaries between them. Two cell types are seen on a slide. Chief cells (also called principal cells) are the small, numerous cells that make up most of the gland. They are the hormone-making cells of the parathyroid. Oxyphil cells are larger and much fewer. Their cytoplasm is packed with mitochondria, so it stains dark pink. Oxyphil cells become more common with age. Their function is not fully understood.

| Structure | What it is | How to find it |
|---|---|---|
| **Parathyroid glands** | Usually four small, round glands on the posterior surface of the thyroid lobes | Two on the back of each lobe, each in its own capsule |
| **Chief cells** | The main hormone-making cells of the parathyroid glands | Small, numerous cells with round nuclei and pale cytoplasm, making up most of the tissue |
| **Oxyphil cells** | Larger, fewer cells whose function is not fully understood | Larger cells with dark pink, granular cytoplasm, alone or in small groups among the chief cells |

- Memory tip: follicles and colloid mean thyroid. Packed cords of cells with no colloid mean parathyroid. On a parathyroid slide, the small, plentiful cells are chief cells and the big pink ones are oxyphil cells.

### What this means for patients

- The parathyroid glands are tiny and embedded on the back of the thyroid, so during thyroid surgery they can be damaged or removed along with it. A surgeon has to find and spare them.
- Because the thyroid sits just under the skin of the front of the neck, an enlarged thyroid (a goiter) can be seen and felt.

## Competency 131. Adrenal gland zones

**In your course notes:** Module 4 packet, Chapter 6, competency 131 (page 56).

**Dr. Rennie's video for this part:** Endocrine system concept video, chapter Major glands and their roles.

**You should be able to:** Identify the adrenal cortex zones in order and the adrenal medulla with its chromaffin cells.

An adrenal gland, also called a suprarenal gland, sits on the superior pole of each kidney. Both adrenal glands lie on the posterior abdominal wall behind the peritoneum, enclosed with the kidney in fat and the renal fascia. The right gland is shaped like a pyramid. The left gland is shaped like a crescent. Each gland is covered by a connective tissue capsule.

Each adrenal gland is two glands in one. The outer adrenal cortex makes up most of the gland. It is glandular epithelial tissue that develops from mesoderm. The inner adrenal medulla is a small core in the center. It is modified nervous tissue that develops from the neural crest, the same embryonic tissue that forms the sympathetic ganglia. On a slide the cortex looks like layered bands and the medulla looks like a different tissue in the middle.

**Capsule**

The connective tissue covering of the gland.

**Adrenal cortex**

The outer region and the bulk of the gland. Glandular epithelial tissue arranged in three zones.

**Adrenal medulla**

The inner core. Modified nervous tissue that belongs to the sympathetic nervous system.

### The three zones of the cortex, from the capsule inward

The cortex is built in three zones. Each zone is named for how its cells are arranged. Follow them in order, from the capsule inward to the medulla.

1. Zona glomerulosa: the thin outer zone, just under the capsule. Its cells form small rounded clusters and arches. "Glomerulus" means a little ball.
2. Zona fasciculata: the thick middle zone, the widest of the three. Its cells line up in long, straight columns running toward the center. "Fascicle" means a bundle. The cells look pale and foamy because they hold many lipid droplets.
3. Zona reticularis: the thin inner zone, against the medulla. Its cells form a branching network. "Reticulum" means a network. These cells stain darker than the fasciculata.

### The adrenal medulla and chromaffin cells

The medulla is made of chromaffin cells. These are large, round to many-sided cells arranged in clusters around blood vessels. Chromaffin cells are modified sympathetic neurons. They have no axons or dendrites, and they release their product straight into the blood. They are supplied by preganglionic sympathetic nerve fibers, so the medulla works like a sympathetic ganglion. Their name comes from the brown color they turn when stained with chromium salts.

| Layer | Part of | Position and thickness | Tissue type | How the cells are arranged |
|---|---|---|---|---|
| **Capsule** | Covering | Outermost, thin | Dense connective tissue | A connective tissue sheet |
| **Zona glomerulosa** | Cortex | Outer, thin | Glandular epithelial tissue | Rounded clusters and arches |
| **Zona fasciculata** | Cortex | Middle, the thickest of the three | Glandular epithelial tissue | Long, straight columns of pale, foamy cells |
| **Zona reticularis** | Cortex | Inner, against the medulla, thinner than the fasciculata | Glandular epithelial tissue | A branching network of darker cells |
| **Adrenal medulla** | Medulla | The core | Modified nervous tissue | Clusters of large chromaffin cells around blood vessels |

- Memory tip: GFR, from the outside in. Glomerulosa, Fasciculata, Reticularis. Then the medulla, full of chromaffin cells, in the center.
- The middle zone is the thickest. If your drawing has the glomerulosa as the widest band, the order or the widths are wrong.

- The three cortex zones are all glandular tissue. The medulla is the only part that is nervous tissue.

### What this means for patients

- The adrenal gland rests on the top of the kidney, so imaging ordered for the kidney usually shows the adrenal as well. That is how an adrenal mass is sometimes found by accident.

## Competency 132. Pancreas, gonads, and organs with endocrine tissue

**In your course notes:** Module 4 packet, Chapter 6, competency 132 (page 58).

**Dr. Rennie's video for this part:** Endocrine system concept video, chapter Pancreas hormones and blood sugar.

**You should be able to:** Locate the pancreatic islets and gonads and name other organs such as the heart, kidneys, and skin that contain endocrine tissue.

### The pancreas and its islets

The pancreas is a long, soft gland in the upper abdomen. It lies behind the stomach, against the posterior abdominal wall and mostly behind the peritoneum. Its head sits in the C-shaped curve of the duodenum. Its body crosses behind the stomach, and its tail reaches toward the spleen on the left.

The pancreas is a mixed gland, with both exocrine and endocrine tissue. Most of it is exocrine tissue. This tissue is made of acini, clusters of cells that make digestive enzymes and send them through ducts to the duodenum. Scattered among the acini are about 1 to 2 million small islands of endocrine cells called the pancreatic islets, or islets of Langerhans. The islets have no ducts. They release their hormones into the many capillaries that run through them. On a slide, an islet is a round, pale-staining patch of cells surrounded by darker-staining acini.

### The gonads

The gonads are mixed organs too. They make gametes, and they also contain endocrine cells. The ovaries are the paired female gonads. They sit in the pelvic cavity, one on each side of the uterus, against the lateral pelvic wall. Their endocrine tissue is the cells of the developing ovarian follicles and the corpus luteum that forms after ovulation. The testes are the paired male gonads. They sit in the scrotum, outside the abdominopelvic cavity. Their endocrine tissue is the interstitial cells (Leydig cells), which lie in the spaces between the seminiferous tubules.

| Structure | Location | Endocrine part |
|---|---|---|
| **Pancreas** | Upper abdomen behind the stomach; head in the curve of the duodenum, tail toward the spleen | Both exocrine and endocrine; a mixed gland |
| **Pancreatic islets** | Scattered throughout the pancreas | The endocrine cell clusters (islets of Langerhans) |
| **Acini** | Most of the pancreas | None. The exocrine clusters that make digestive enzymes |
| **Ovaries** | Pelvic cavity, one on each side of the uterus | Cells of the ovarian follicles and the corpus luteum |
| **Testes** | Scrotum | Interstitial (Leydig) cells between the seminiferous tubules |

- The islets are the endocrine part of the pancreas and the acini are the exocrine part. Islets are pale, acini are dark, and the islets are the small patches.
- These notes call three organs mixed organs: the pancreas, the ovaries, and the testes. An organ is mixed when it does an endocrine job alongside another role, such as making digestive enzymes or making gametes.

### Other organs with endocrine tissue

Many organs whose main job is something else still contain endocrine cells. These organs are not counted as endocrine glands, and they are not called mixed organs either, but their endocrine cells still release hormones into the blood. The heart has hormone-making cells in the muscle of the atrial walls. The kidneys have endocrine cells in the renal cortex. The

skin takes part too: cells of the epidermis begin making a hormone when the skin is exposed to sunlight. The stomach and small intestine, the liver, adipose tissue, and the placenta during pregnancy also contain endocrine cells.

| Organ | Where it sits | Where the endocrine cells are |
|---|---|---|
| **Hypothalamus** | Brain, floor and walls of the third ventricle | Neurosecretory cells. A neuroendocrine organ that controls the pituitary gland |
| **Heart** | Chest, in the mediastinum | Cardiac muscle cells in the walls of the atria |
| **Kidneys** | Abdomen, against the posterior wall | Endocrine cells in the renal cortex, including the juxtaglomerular cells in the walls of the afferent arterioles |
| **Skin** | Body surface | Keratinocytes of the epidermis, which begin making a hormone when exposed to sunlight |
| **Stomach and small intestine** | Abdomen | Enteroendocrine cells scattered in the mucosa |
| **Liver** | Upper right abdomen | Hepatocytes, the main liver cells |
| **Adipose tissue** | Under the skin and around organs | Adipocytes (fat cells) |
| **Placenta** | In the uterus during pregnancy | Placental cells. A temporary organ, because it exists only during pregnancy |

- The heart, kidneys, and skin contain endocrine tissue, but they are not endocrine glands and these notes do not count them as mixed organs. Only the pancreas, ovaries, and testes get that label.
- The placenta is the one temporary organ on the list. It forms during pregnancy and is gone afterward.

## Worked questions

Nine questions in the style of the lecture exam, at least one for each competency. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 128

**Which endocrine gland sits in the sella turcica of the sphenoid bone?**

A. Pineal gland

B. Hypothalamus

C. Pituitary gland

D. Thyroid gland

**Answer: C.** The pituitary is the pea-sized gland that sits in the sella turcica, the saddle-shaped pocket of the sphenoid bone, hanging below the hypothalamus by the infundibulum.

**Why the others are wrong**

- **A** The pineal gland is in the epithalamus, projecting from the posterior roof of the third ventricle. It is in the brain, not in a pocket of bone.
- **B** The hypothalamus is the floor and lower walls of the third ventricle, above the pituitary. It is brain tissue, not a gland in the sella.
- **D** The thyroid is in the anterior neck, below the larynx, with its lobes beside the trachea.

QUESTION 2 · COMPETENCY 128

**Which gland is the largest purely endocrine gland in the body?**

A. Thyroid gland

B. Pancreas

C. Thymus

D. Adrenal gland

**Answer: A.** The thyroid, the butterfly-shaped gland in the front of the neck, is the largest gland that is entirely endocrine.

**Why the others are wrong**

- **B** The pancreas is larger, but it is a mixed gland. Most of it is exocrine acini, and only the scattered islets are endocrine.
- **C** The thymus in the mediastinum is large in childhood but shrinks after puberty. It is not the largest purely endocrine gland.
- **D** Each adrenal gland is a small gland capping the top of a kidney, much smaller than the thyroid.

QUESTION 3 · COMPETENCY 129

**The posterior lobe of the pituitary (neurohypophysis) is made of which tissue?**

A. Glandular epithelial tissue

B. Dense connective tissue

C. Simple cuboidal epithelium

D. Nervous tissue

**Answer: D.** The posterior lobe grows down from the floor of the brain and is an extension of the hypothalamus. It holds the axons of hypothalamic neurons and supporting pituicytes. "Neuro" means nervous tissue.

**Why the others are wrong**

- **A** Glandular epithelial tissue is the anterior lobe, the adenohypophysis, which grows up from the roof of the embryonic mouth.
- **B** Dense connective tissue forms capsules such as the one around the adrenal gland, not a pituitary lobe.
- **C** Simple cuboidal epithelium is the follicular cell layer of the thyroid follicle, not pituitary tissue.

QUESTION 4 · COMPETENCY 129

**How does the anterior lobe of the pituitary receive its signals from the hypothalamus?**

A. Through the hypothalamo-hypophyseal tract of nerve fibers

B. Through the blood vessels of the hypophyseal portal system

C. Through the axons of pituicytes

D. Through the optic chiasm

**Answer: B.** The anterior lobe is glandular epithelium with no nerve link to the hypothalamus. Hypothalamic signals enter the primary capillary plexus, travel down the hypophyseal portal veins along the infundibulum, and reach the gland cells at the secondary capillary plexus.

**Why the others are wrong**

- **A** The hypothalamo-hypophyseal tract is the nerve fiber bundle to the posterior lobe, the nervous-tissue lobe.
- **C** Pituicytes are the supporting glial cells of the posterior lobe. They do not carry signals to the anterior lobe.
- **D** The optic chiasm lies just above and in front of the pituitary. A pituitary tumor presses on it, but it carries no signals to the gland.

QUESTION 5 · COMPETENCY 130

**On a thyroid slide, where is the colloid?**

A. Filling the center of each follicle

B. Between the follicles

C. In the cytoplasm of the parafollicular cells

D. In the connective tissue capsule

**Answer: A.** Each thyroid follicle is a ring of follicular cells around a center filled with colloid, the thick, protein-rich fluid that stains pink.

**Why the others are wrong**

- **B** Between the follicles are the parafollicular (C) cells and capillaries, not colloid. The C cells never touch the colloid.
- **C** Parafollicular cells are larger, paler cells that sit outside the follicle wall. Colloid is stored inside the follicle, not inside any cell.
- **D** The capsule is a thin connective tissue covering on the outside of the gland.

QUESTION 6 · COMPETENCY 130

**Which cells make up most of a parathyroid gland and are its hormone-making cells?**

A. Follicular cells

B. Oxyphil cells

C. Chief cells

D. Parafollicular cells

**Answer: C.** Chief cells (principal cells) are the small, numerous cells with round nuclei and pale cytoplasm that make up most of the parathyroid tissue, and they are the gland's hormone-making cells.

**Why the others are wrong**

- **A** Follicular cells form the wall of a thyroid follicle. They are thyroid cells, not parathyroid cells.
- **B** Oxyphil cells are the larger, fewer parathyroid cells with dark pink cytoplasm. They become more common with age and their function is not fully understood.
- **D** Parafollicular (C) cells are thyroid cells sitting between the follicles.

QUESTION 7 · COMPETENCY 131

**Which list gives the zones of the adrenal cortex in order from the capsule inward?**

A. Zona reticularis, zona fasciculata, zona glomerulosa

B. Zona glomerulosa, zona fasciculata, zona reticularis

C. Zona fasciculata, zona glomerulosa, zona reticularis

D. Zona glomerulosa, zona reticularis, zona fasciculata

**Answer: B.** GFR from the outside in: the thin glomerulosa under the capsule, the thick fasciculata in the middle, and the thin reticularis against the medulla.

**Why the others are wrong**

- **A** That is the order from the medulla outward, backwards from the capsule.
- **C** The glomerulosa is the outer zone, just under the capsule. The fasciculata is the middle zone, not the outer one.
- **D** The reticularis is the inner zone against the medulla. The fasciculata sits between the glomerulosa and the reticularis.

QUESTION 8 · COMPETENCY 131

**Chromaffin cells are found in which part of the adrenal gland?**

A. Zona fasciculata

B. Capsule

C. Zona glomerulosa

D. Adrenal medulla

**Answer: D.** The medulla, the core of the gland, is made of chromaffin cells. They are modified sympathetic neurons with no axons or dendrites, clustered around blood vessels, and they release their product straight into the blood.

**Why the others are wrong**

- **A** The zona fasciculata is the thick middle cortex zone, made of pale, foamy gland cells in straight columns.
- **B** The capsule is dense connective tissue covering the outside of the gland.
- **C** The zona glomerulosa is the thin outer cortex zone, made of gland cells in rounded clusters and arches.

QUESTION 9 · COMPETENCY 132

**Which organ contains endocrine tissue but is not counted as an endocrine gland or a mixed organ in these notes?**

A. Pancreas

B. Heart

C. Testis

D. Thyroid gland

**Answer: B.** The heart's main job is pumping blood, but cardiac muscle cells in the walls of the atria also release a hormone. It contains endocrine tissue without being an endocrine gland, and these notes do not call it a mixed organ.

**Why the others are wrong**

- **A** The pancreas is one of the three mixed organs in these notes: exocrine acini plus endocrine islets.
- **C** The testis is a mixed organ too. It makes gametes and has endocrine interstitial (Leydig) cells between the seminiferous tubules.
- **D** The thyroid is a purely endocrine gland, the largest one in the body.